Q2 Review Report: Speaker Classification

Two models are implemented:

- **Baseline model using MFCC features**
- **Improved model using enriched acoustic features**

Dataset

- **Dataset: LibriSpeech (train-clean-100)**
- **Audio format: .flac**
- **Number of speakers: multiple**
- **Samples used: 500 (subset)**

The dataset provides high-quality speech recordings suitable for speaker classification tasks.

Methodology

Feature Extraction

Baseline Features:

- **MFCC (13 coefficients)**
- **Captures spectral characteristics of speech**

Improved Features:

- **MFCC (13)**
- **Energy (signal power)**
- **Zero Crossing Rate (ZCR)**

These additional features capture:

- **Signal intensity (energy)**
- **Frequency changes (ZCR)**

Model Architecture

Both models use a simple neural network:

- **Input Layer**
- **Fully Connected Layer (64 units)**
- **ReLU Activation**
- **Output Layer (number of speakers)**

Loss Function:

- **CrossEntropyLoss**

Optimizer:

- **Adam (learning rate = 0.001)**

Epochs:

- **20**

Results

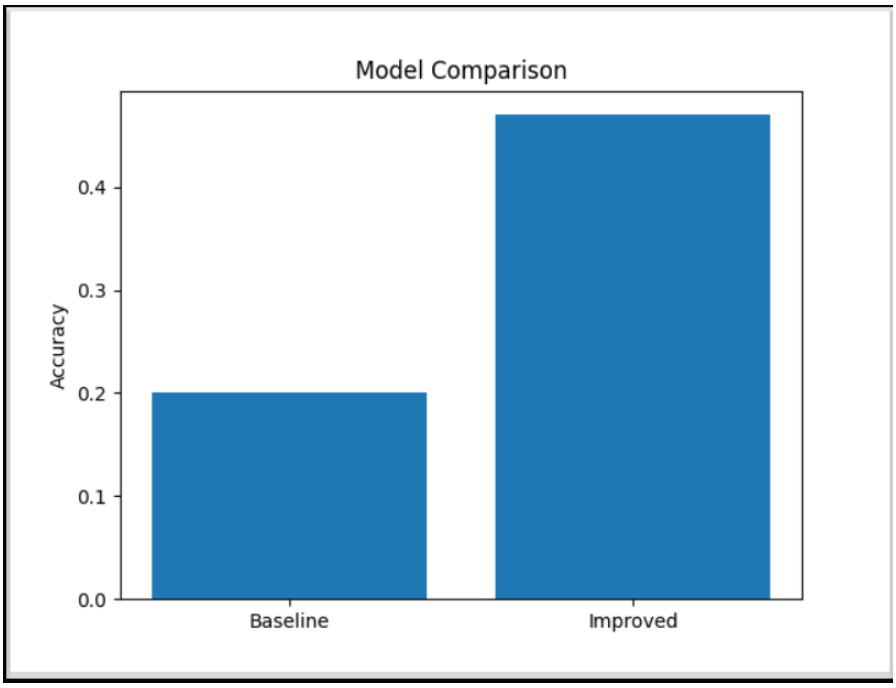
Model	Features	Accuracy
Baseline	MFCC	0.20
Improved	MFCC + Energy + ZCR	0.47

Analysis

- **The baseline model achieves low accuracy (0.20), indicating that MFCC alone is insufficient for speaker discrimination.**
- **The improved model significantly increases accuracy to 0.47.**
- **This demonstrates that adding simple acoustic features enhances model performance.**

Visualization

A comparison plot was generated:



Discussion

- **Feature engineering plays a crucial role in speech tasks.**
- **Even simple additions like energy and ZCR improve classification.**
- **The model is limited by:**
    - **Small dataset subset**
    - **Simple architecture**

Future improvements:

- **Use full dataset**
- **Apply deep learning models (CNN/RNN)**
- **Use embeddings (e.g., wav2vec)**
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Conclusion

This experiment demonstrates that:

- **MFCC features provide a baseline for speech representation**
- **Additional acoustic features significantly improve performance**
- **Proper feature design is essential for effective speaker classification**

The improved model outperforms the baseline, validating the importance of richer feature representation.